INCH-POUND
MIL-DTL-24643/7E
22 August 2002
SUPERSEDING
MIL-C-24643/7D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size No. 16, class K stranding, uncoated.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth - The specified number of conductors (see table I) cabled together in one or more layers, all in left hand direction, with the minimum amount of fillers necessary, to obtain a firm well-rounded assembly. Cabling sequence to be consecutive, starting with No. 1, from center outward. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. A single conductor shall not be used as center for the cable core.
Fifth - Binder tape applied helically with overlap.
Sixth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:55Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/7	Type and size	Number of conductors	Overall diameter		Bending endurance roller diameter (maximum) (inches)
			min (inches)	max (inches)	
-01UN	LSMHOF-7	7	0.465	0.500	2.50
-02UN	LSMHOF-10	10	0.540	0.585	2.50
-03UN	LSMHOF-14	14	0.585	0.635	3.00
-04UN	LSMHOF-19	19	0.650	0.705	3.00
-05UN	LSMHOF-24	24	0.735	0.795	3.50
-06UN	LSMHOF-30	30	0.775	0.835	4.00
-07UN	LSMHOF-37	37	0.855	0.925	4.00
-08UN	LSMHOF-44	44	0.925	1.000	4.50
-09UN	LSMHOF-61	61	1.100	1.175	5.00

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 4.59
- Voltage withstand - volts, root mean square, minimum 2000
- Insulation resistance - megohms -1000 feet, minimum 500
- Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C) 50
  - Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum. 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2,minimum 1800
    - Elongation - percent, minimum 250
  - Jacket (cable)
    - Tensile strength - lb/in^2, minimum 1300
    - Elongation - percent, minimum 160
  - Twisting and bending endurance -
    - At 75°C, cycles, minimum 6000
    - At minus 30 ± 2°C, cycles, minimum 2000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:55Z
Check the source to verify that this is the current version before use.

Group C:
Physicals (aged)
- Insulation
  - Ethylene propylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
  - Jacket (cable)
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Permanence of printing (jacket) - cycles, minimum 125
- Permanence of printing (conductor, method 1 only) - cycles, minimum 25
- Heat distortion - percent of unaged, maximum 30

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

- Aging and compatibility (cable) (125 ± 5°C) ..... No failure
- Abrasion resistance (jacket) - scrapes, minimum 75
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
- Halogen content - percent, maximum
  - Jacket 0.2
  - Fillers 0.2
- Immersion (jacket)
  - Tensile strength - percent of unaged, minimum 50
  - Elongation - percent of unaged, minimum 50
- Smoke index, maximum
  - Jacket 25
  - Fillers 45

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:55Z
Check the source to verify that this is the current version before use.

Insulation 45
Toxicity index, maximum
Jacket 5
Fillers 5
Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTH:
All sizes 1500 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-005)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:55Z
Check the source to verify that this is the current version before use.